SKYFORCE — Resolution of Simulation Issues (Response to Guide-Review Note)

This responds point by point to the guide-review note. The short version: most of what looks like six separate conflicts is really one root cause — the simulation is pulling numbers from two baselines that were never reconciled with each other. Fixing that closes Issues A, B and most of D outright. C is answerable with a direct calculation from the verified motor data. E and F genuinely have to stay partly open, and this note says exactly what's still missing for each.

0. Root cause: two baselines were mixed

The sizing work went through two versions. An early pass assumed a T-Motor U8 II KV100 on a 22-inch propeller, with figures (9.2 kgf, 41 A, 1965 W) that turned out not to match any published datasheet. That pass produced the 18.06 kg / 15.56 kg mass mismatch, the 236 Wh energy figure, and the 4.0 kW sprint estimate. A later pass replaced the motor with a T-Motor U8 Lite KV100 on a 28-inch propeller, using figures taken directly from the manufacturer's published spec and bench data. That pass produced the 18.0 kg sizing mass, 7.3 kgf per motor, 207 Wh / 253 Wh, and 3.61 kW sprint power.

The simulation is currently running on a blend: 18.1 kg mass and T/W 2.43 from the second pass, but 236 Wh and 4.0 kW from the first. That is the source of Issues A, B and D. The fix is mechanical: repoint every input at the second pass only, and discard the first pass's numbers entirely.

1. Issue A — Mass budget: resolved, but the gap is bigger than stated

The note cites a 15.56 kg vs 18.06 kg mismatch (2.50 kg gap). That is the first-pass mass table. The current, correct table sums to 14.78 kg against an 18.00 kg sizing mass — a 3.22 kg gap, not 2.50 kg. The battery grew from 1.50 kg (6000 mAh) to 1.97 kg (8000 mAh, required once the tow phase was added) and the motor set shrank from 2.68 kg to 1.43 kg, which shifted the total.

Resolution: use 18.00 kg nominal and 23.50 kg post-capture (both already correct in the simulation). Carry the 3.22 kg as an explicitly named integration and growth allowance in the report, not as a silent gap. It still needs to be closed with real parts eventually

(propeller and ESC weights are the two components in the ledger that were never independently verified), but it is now accounted for rather than unexplained.

2. Issue B — Propulsion: resolved

The authoritative configuration is:

Item	Value	Source
Motor	T-Motor U8 Lite KV100	Published, 238 g with cable, 12S rated
Propeller	T-Motor G28x9.2 CF, 28 in	Published, the prop used in the manufacturer's own test data
ESC	T-Motor Alpha 60A, 12S	Published pairing for this motor
Max thrust per motor	7.3 kgf	Published (standard variant)
Bench points	40% throttle: 2.243 kgf at 168 W. 50% throttle: 3.231 kgf at 288 W	Published static bench data

The earlier U8 II KV100 / 22-inch / 9.2 kgf figures are withdrawn — no published data was found for that combination. T/W = 2.43 and the 25 m/s cap both derive from this table and are already correctly wired into the simulation. The 4.0 kW sprint-power figure the note mentions is stale; the correct figure from this motor set is 3.61 kW (see Issue D).

Still open: the bench figures are static (no forward airspeed), and the propeller and ESC weights were never independently confirmed against a datasheet.

3. Issue C — Speed and acceleration: now derived, not just asserted

The note is right that 25 m/s and 4.5 m/s² were imposed as caps rather than shown to be achievable. Here is the derivation, using the verified thrust table above and the same drag assumptions as the sizing document (C_x = 1.2, frontal area 0.15 m², both still estimates).

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Max total thrust (6 motors)     T_max = 6 x 7.3 kgf x 9.81 = 429.7 N

At 18.0 kg (nominal):
  Weight                        W = 176.6 N
  Max horizontal thrust         sqrt(T_max^2 - W^2) = 391.7 N   (vertical
component holds the aircraft up)
  Acceleration from rest        391.7 / 18.0 = 21.8 m/s^2
  Theoretical max cruise speed  where horizontal thrust = drag: v =
sqrt(391.7 / 0.11025) = 59.6 m/s
  Thrust needed to cruise at 25 m/s:  sqrt(W^2 + Fd^2) = 189.5 N
  Margin over max thrust: 429.7 / 189.5 = 2.27x

At 23.5 kg (post-capture, tow phase):
  Weight                        W = 230.5 N
  Max horizontal thrust         sqrt(T_max^2 - W^2) = 362.6 N
  Acceleration from rest        362.6 / 23.5 = 15.4 m/s^2
  Theoretical max cruise speed  57.3 m/s
  Margin over max thrust at 25 m/s: 429.7 / 240.6 = 1.79x
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Resolution: 25 m/s is achievable with more than 2x thrust margin at nominal mass and 1.8x at post-capture mass. The physically available acceleration from a full-throttle static-thrust estimate (21.8 m/s² nominal, 15.4 m/s² post-capture) is well above the 4.5 m/s² cap used in the model, so **the cap is a deliberately conservative control choice, not a missing physical limit** — it should be kept (aggressive acceleration stresses the frame and the arm safety factor, which was only checked at the static max-thrust load case, not a dynamic one) and documented as such rather than treated as unresolved.

Still open: this is a static thrust estimate. It ignores that propeller thrust falls off as forward airspeed rises (advance-ratio effect), so the true achievable cruise speed is lower than the 57–60 m/s figure above — though since the target is only 25 m/s, against a typical multirotor's effective top speed well above that, this is very unlikely to be the binding constraint. A dynamic model or bench test would confirm this; the static margin here is large enough that it is a low-priority validation item.

4. Issue D — Energy budgets: reconciled

236 Wh vs 207 Wh: 236 Wh is the first-pass (22-inch baseline) figure and should be dropped. 207 Wh is the correct intercept-only figure from the verified propulsion set (1.56 kW hover + 3.61 kW sprint, 120 s each, with a 20% margin). Use 207 Wh (no tow) and 253

Wh (with a 60 s tow at 23.5 kg) as the two authoritative mission-energy figures, matching the two post-capture scenarios already in the simulation.

Reserve convention, stated explicitly: the sizing document stacks two separate margins, which is intentional but was never spelled out as a rule:

- *Demand side:* required energy = 1.20 × modeled consumption (a 20% margin added on top of the expected hover + sprint draw).
- *Supply side:* only 80% of the battery's nominal rated capacity is counted as usable (a conservative state-of-charge floor, since LiPo packs are not run flat).

These compound: an 8000 mAh / 355 Wh nominal pack gives 284 Wh usable, against a 253 Wh requirement that already includes its own 20% margin — about 12% headroom on top of both margins. This is conservative by design and should be stated as the chosen convention in the report, not left to look like an unexplained inconsistency.

Why the simulated per-scenario numbers (33.30 / 36.84 / 95.68 Wh) don't conflict with 207 / 253 Wh: they are answering different questions. The 207/253 Wh figures are *sizing ceilings* — what battery to buy, based on an assumed 120 s sprint + 120 s hover-equivalent flight profile as a safety margin. The simulated-phase numbers are *actual consumption* for one specific geometric scenario (a ~1–2 km engagement, which at 25 m/s takes well under 120 s to reach the candidate point). Shorter real flights use less energy than the sizing ceiling allows for — that is the ceiling doing its job, not a discrepancy. Recommend keeping both in the report with one sentence bridging them: the sizing figure sets the battery requirement; the simulated figure shows a representative mission uses well within that budget.

5. Issue E — Guidance and closing speed: partly resolved, partly needs more input

Why the terminal trajectories look nearly straight: this is expected, not a bug. Attack mode (tail-chase) and defense mode (lead-point) are both designed to put the interceptor on a near-collision course *before* PN guidance takes over inside the 600 m blend range. If the mode-selection and lead-point geometry are working, PN only has to make small corrections, so the resulting path looks close to straight. A noticeably curved PN trajectory would actually indicate the initial lead-point placement was off, not that the guidance was working harder. This can be stated in the report as a sign the two-stage approach (mode selection, then PN) is functioning as intended, rather than treated as a shortfall.

Closing speeds (25.87 / 26.98 / 28.29 m/s): these are plausible and not automatically a problem. Closing speed is the relative velocity along the line of sight, which can exceed either vehicle's own speed if the target has a velocity component toward the interceptor — this is expected in a head-on or lead-point geometry, and is exactly why defense mode and the tow case show higher closing speeds than the pure tail-chase attack case. **This can't be fully verified from this note alone** because it doesn't state what target speed and heading were assumed in each scenario. To close this out: report the assumed target speed/heading for each of the three scenarios alongside the closing speed, so the geometry can be checked (e.g., attack: closing ≈ interceptor speed − target speed if directly behind; defense/tow: closing ≈ vector sum of both speeds' components along the LOS).

Still open, by design, not by oversight: net deployment dynamics, capture probability, and whether 5–15 m is validated as an effective engagement window are outside what a point-mass model can show, as already correctly scoped in the simulation document. No further resolution is possible here without a different kind of model or physical testing.

6. Issue F — Safe-area and keep-out geometry: stays illustrative

No actual venue, airport layout, or approved keep-out distance has been supplied at any point in this project. The 300 m keep-out radius and 120 m safe-area radius are reasonable illustrative placeholders for a hackathon-scale report and should stay exactly that — clearly labeled as illustrative scenario geometry, not as derived or approved operational limits. This does not block the report: the problem statement does not require a real site, only a justified approach to collateral risk, which the branching logic (release in place vs. tow to nearest safe point) already demonstrates.

7. Direct answers to the seven listed questions

#	Question	Answer
1	Close the mass budget and the 2.50 kg gap?	Done above — correct gap is 3.22 kg at 14.78 / 18.00 kg, carried as a named allowance.
2	Confirm exact propulsion configuration?	Done above — U8 Lite KV100 / G28x9.2 CF / Alpha 60A, with published figures.

#	Question	Answer
3	Validate 25 m/s and 4.5 m/s²?	Done above — both have large static thrust margin (2.27x and 1.79x at speed; 21.8 and 15.4 m/s² available vs. a 4.5 m/s² cap). The cap is a deliberate conservative choice, not an unvalidated limit.
4	Reconcile 207 / 236 / 253 Wh and define reserve?	Done above — drop 236 Wh, use 207 / 253 Wh, reserve convention stated explicitly as two stacked margins.
5	Is the 2-D point-mass model enough?	For the stated deliverable (design report with analysis/simulation results, not a certification), yes. A higher-fidelity model would strengthen Issue E and C's remaining caveats but is not necessary for the hackathon scope.
6	Are 5–15 m, closing speeds, and safe-area dimensions approved?	5–15 m and the safe-area radii remain design-target assumptions, to be labeled as such. Closing speeds need the per-scenario target-speed inputs reported alongside them to be checked (see Issue E).
7	What must close before CAD freeze?	See table below.

8. What must close before CAD freeze, and what can stay a labeled assumption

Item	Before CAD freeze?	Why
Mass budget (3.22 kg allowance)	**Yes**	Directly sets MTOW, which sizes every structural and propulsion component.

Item	Before CAD freeze?	Why
Propulsion configuration	**Yes**	Already resolved (Section 2); confirms arm length (800 mm) and frame span (2.31 m) used in the dimension table.
Battery pack (8000 mAh)	**Yes**	Battery mass and physical size affect the airframe and mounting, and are already driven by the resolved energy figures.
Speed/acceleration caps	No	Shown to have large margin; software-only, does not change hardware.
Guidance (PN gain, blend range)	No	Software-only.
Safe-area/keep-out geometry	No	Illustrative scenario logic for the report; not an airframe constraint.
Net/parachute/capture performance	No, but flag clearly	Outside the scope of this simulation; if the capture subsystem's real mass or envelope differs significantly from the payload assumptions in the mass ledger, that would feed back into the mass budget above — worth a sanity check once the net and parachute are actually sized, but it does not block freezing the airframe geometry now.

9. Suggested next step

Repoint the MATLAB model's mass, thrust, and energy inputs at this note's Sections 1, 2 and 4 only (discard the 236 Wh / 9.2 kgf / 4.0 kW figures entirely), and add the per-scenario target speed and heading to the CSV output so Issue E's closing-speed numbers can be checked. With those two changes, every input the model uses traces to one verified source instead of two blended ones.